ICT Lab 11

Tasks

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Course: Information and Communication Technology
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Task 1:

```
#include <iostream>

using namespace std;


class Person

{

public:

	string name;

	int age;

	void display()

	{

	cout << "\n\nName: " << name << "\nAge: "<< age << endl;

	}

};


class student : public Person

{

public:

	int RollNumber;

	void showRoll()

	{

	cout << "Roll Number: " << RollNumber << endl;

	}

};


int main()
```

```
{
        student Stud;
        cout << "\nEnter Student Name: " << "\n";
        cin >> Stud.name;
        cout << "\nEnter Student Age: " << "\n";
        cin >> Stud.age;
        cout << "\nEnter Student Roll Number: " << "\n";
        cin >> Stud.RollNumber;
        Stud.display();
        Stud.showRoll();
return 0;
}
```

ICTLab11Task1.cpp | ICTLab11Task2.cpp

```cpp
#include <iostream>
using namespace std;

class Person
{
public:
    string name;
    int age;
    void display()
    {
    cout << "\n\nName: " << name << "\nAge: "<< age << endl;
    }
};

class student : public Person
{
public:
    int RollNumber;
    void showRoll()
    {
    cout << "Roll Number: " << RollNumber << endl;
    }
};

int main()
{
    student Stud;
    cout << "\nEnter Student Name: " << "\n";
    cin >> Stud.name;
    cout << "\nEnter Student Age: " << "\n";
    cin >> Stud.age;
    cout << "\nEnter Student Roll Number: " << "\n";
    cin >> Stud.RollNumber;
    Stud.display();
    Stud.showRoll();
return 0;
}
```

```
Enter Student Name:
Farhan

Enter Student Age:
19

Enter Student Roll Number:
0538


Name: Farhan
Age: 19
Roll Number: 538
[Finished in 7.85s]
```

Task 2:

```
#include <iostream>

using namespace std;

class Employee

{

public:

	void work()

	{

	cout << "\nEmployee is working"<< endl;

	}

};


class Manager : public Employee

{

public:

	void work()

	{

	cout << "\nManager is managing work"<< endl;

	}

};


int main()

{

	Employee Emp;

	Manager Manag;
```

```
        Emp.work();
        Manag.work();
return 0;
}
```

ICTLab11Task1.cpp ICTLab11Task2.cpp

```cpp
#include <iostream>
using namespace std;
class Employee
{
public:
    void work()
    {
    cout << "\nEmployee is working"<< endl;
    }
};

class Manager : public Employee
{
public:
    void work()
    {
    cout << "\nManager is managing work"<< endl;
    }
};

int main()
{
    Employee Emp;
    Manager Manag;

    Emp.work();
    Manag.work();
return 0;
}
```

```
Employee is working

Manager is managing work
[Finished in 0.46s]
```

Task 3:

```
#include <iostream>

using namespace std;


class Employees

{

public:

                string name;

                int salary;

                void virtual calculateBonus()

                {

                  cout << "\nBase Employee bonus is 1000." << endl;

                }

};


class Manager : public Employees

{

public:

                void virtual calculateBonus()

                {

                  cout << "\nManager bonus is 5000." << endl;

                }

};


class Developer : public Employees

{
```

```
public:
        void virtual calculateBonus()
        {
          cout << "\nDeveloper bonus is 3000." << endl;
        }
};

int main()
{
        Employees* emps[2];
        Developer dev;
        Manager manag;
        emps[0] = &manag;
        emps[1] = &dev;

        for (int i = 0; i < 2; ++i)
        {
          cout << "\nEnter Employee#" << i+1 << " Name: " << endl;
          cin >> emps[i]->name;
          cout << "\nEnter Employee#" << i+1 << " salary: " << endl;
          cin >> emps[i]->salary;
        }

        for (int i = 0; i < 2; ++i)
        {
          cout << "\nEnter Employee#" << i+1 << " Name: " << emps[i]->name << endl;
```

```
            cout << "\nEnter Employee#" << i+1 << " Salary: " << emps[i]->salary << endl;
            emps[i] -> calculateBonus();
        }
return 0;
}
```

ICTLab11Task1.cpp ICTLab11Task2.cpp ICTLab11Task3.cpp

```
{
public:
    void virtual calculateBonus()
    {
        cout << "\nManager bonus is 5000." << endl;
    }
};

class Developer : public Employees
{
public:
    void virtual calculateBonus()
    {
        cout << "\nDeveloper bonus is 3000." << endl;
    }
};

int main()
{
    Employees* emps[2];
    Developer dev;
    Manager manag;
    emps[0] = &manag;
    emps[1] = &dev;

    for (int i = 0; i < 2; ++i)
    {
        cout << "\nEnter Employee#" << i+1 << " Name: " << endl;
        cin >> emps[i]->name;
        cout << "\nEnter Employee#" << i+1 << " salary: " << endl;
        cin >> emps[i]->salary;
    }

    for (int i = 0; i < 2; ++i)
    {
        cout << "\nEnter Employee#" << i+1 << " Name: " << emps[i]->name << endl;
        cout << "\nEnter Employee#" << i+1 << " Salary: " << emps[i]->salary << endl;
        emps[i] -> calculateBonus();
    }
return 0;
}
```

```
Enter Employee#1 Name:
Farhan

Enter Employee#1 salary:
100000

Enter Employee#2 Name:
Mohsin

Enter Employee#2 salary:
120000

Enter Employee#1 Name: Farhan

Enter Employee#1 Salary: 100000

Manager bonus is 5000.

Enter Employee#2 Name: Mohsin

Enter Employee#2 Salary: 120000

Developer bonus is 3000.
[Finished in 16.89s]
```

Unrelated Note: The Text Editor used is “Sublime Text”. Rather than Dev C++ or VS Code.

Its similar to Dev C++ but very modern, updated and much faster, its also known as the fastest text editor in the entire market. Its specialty is that it can even open huge files and codes almost instantly and without lagging.